

Rigidity of Rationally Connected Smooth Projective Varieties from Dynamical Viewpoints

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Background & Motivation

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- (Nakayama and Zhang, 2009) Building blocks of étale endomorphisms of complex projective manifolds.
(Iitaka fibration, Albanese map and MRC fibration)
- (Meng and Zhang, 2018) EMMP for polarized endomorphisms.
(Abelian varieties & Rationally connected varieties)
- (Meng, 2020) EMMP for int-amplified endomorphisms.
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Definition

- 1 f is polarized if $f^*H \sim qH$ for some ample divisor H and $q > 1$.
- 2 f is int-amplified if $f^*H - H$ is ample for some ample divisor H .

Notation and Early Results

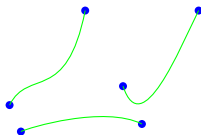
Let X be a normal projective variety of dimension n over $\mathbb{C}$.

- X is **rationally connected**, if any two general points of X can be connected by a rational curve.

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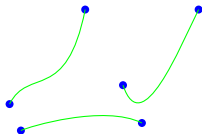
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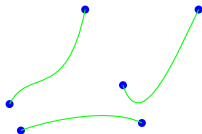


- X is **of Fano type**, if there exists an effective Weil $\mathbb{Q}$ -divisor Δ on X such that the pair (X, Δ) has at worst klt singularities and $-(K_X + \Delta)$ is ample. In particular, if $\Delta = 0$, then X is **Fano**.

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Rationally connected & Fano type

- (Qi Zhang, 2006) F.T. (or more generally, log Q-Fano) $\Rightarrow$ R.C..
- (Yoshikawa, 2020) A smooth rationally connected (SRC) projective variety admitting an int-amplified endomorphism is of Fano type.

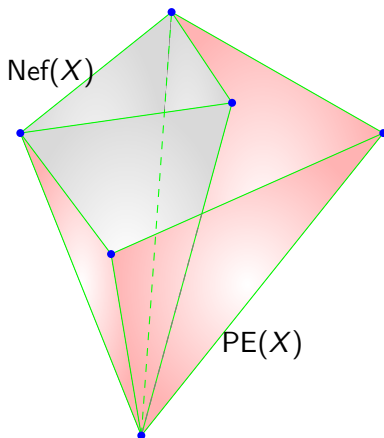
Notation & Early results

- $\text{Pic}(X)$, the group of Cartier divisors of X modulo linear equivalence $\sim$
- $\text{Pic}^\circ(X)$, the neutral connected component of $\text{Pic}(X)$
- $\text{NS}(X) := \text{Pic}(X)/\text{Pic}^\circ(X)$, the Néron-Severi group of X
- $N^1(X) := \text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$, the space of $\mathbb{R}$ -Cartier divisors modulo numerical equivalence $\equiv$
- $N_r(X)$, the space of r -cycles modulo weak numerical equivalence $\equiv_w$

$$x \equiv_w y \iff x \cdot D_1 \cdots D_r = y \cdot D_1 \cdots D_r, \quad \forall D_i \text{ Cartier}$$

- $\rho(X) := \dim_{\mathbb{R}} N^1(X)$, the Picard number of X
- $\text{Nef}(X)$, the cone of nef classes in $N^1(X)$
- $\text{PE}(X)$, the cone of pseudo-effective classes in $N^1(X)$

Notation & Early results



Question

When will $\text{Nef}(X) = \text{PE}(X)$?

Main results

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Theorem (Meng-Z., 2020)

Let X be a SRC projective variety of dimension n . Then

$$X \cong (\mathbb{P}^1)^{\times n}$$

$\iff X$ admits a surjective endomorphism f such that the eigenvalues of $f^|_{N^1(X)}$ (without counting multiplicities) are n distinct real numbers greater than 1.*

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$\iff X$ admits a surjective endomorphism f such that the eigenvalues of $f^|_{N^1(X)}$ (without counting multiplicities) are n distinct real numbers greater than 1.*

Equivalent condition (cf. [Meng, 2020])

*f is int-amplified ($f^*H - H = L$ for some ample divisors H and L) $\iff$ all eigenvalues of $f^*|_{N^1(X)}$ are of modulus greater than 1.*

Examples

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Example (Hirzebruch surface with Picard number 2)

- $X \cong \mathbb{F}_n = \mathbb{P}(\mathcal{O} \oplus \mathcal{O}(-n))$ with $n \geq 1$. Any non-isomorphic surjective endomorphism of X is polarized (eigenvalues are the same modulus).

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Example (Mori-Mukai (1981), Fano threefold of Picard number 3)

- X is a double cover of $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$ with branch locus a divisor of tridegree $(2, 2, 2)$. In particular, $X \not\cong \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$.
- X is smooth Fano but **not rational**; $\text{Nef}(X) = \text{PE}(X)$.
- There are three Fano contractions $\pi_i : X \rightarrow \mathbb{P}^1 \times \mathbb{P}^1$ (conic bundles with degenerate locus non-empty), i.e., each π_i is not a $\mathbb{P}^1$ -bundle.

Theorem (D.-Q. Zhang, 2012)

A smooth Fano threefold admitting a non-isomorphic surjective endomorphism $f : X \rightarrow X$ is rational.

Dynamics with Hodge Index Theorem

Dynamics with Hodge Index Theorem

Lemma (Dinh-Sibony, 2004; D.-Q. Zhang, 2016)

Let X be a normal projective variety. Let $D_1 \neq 0$ and $D_2 \neq 0$ be two nef $\mathbb{R}$ -Cartier divisors such that $D_1 \cdot D_2 \equiv_w 0$. Then $D_1 \equiv aD_2$ for some $a > 0$.

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Let X be a normal projective variety. Let $D_1 \not\equiv 0$ and $D_2 \not\equiv 0$ be two nef $\mathbb{R}$ -Cartier divisors such that $D_1 \cdot D_2 \equiv_w 0$. Then $D_1 \equiv aD_2$ for some $a > 0$.

Proposition (Dinh-Sibony, 2004)

Let $f : X \rightarrow X$ be a surjective endomorphism of a normal projective variety X of dimension n . Let $D, D', D_1, \dots, D_k$ ($k \leq n - 2$) be nef $\mathbb{R}$ -Cartier divisors such that $D \cdot D_1 \cdots D_k \not\equiv_w 0$ and $D' \cdot D_1 \cdots D_k \not\equiv_w 0$. Suppose $f^(D \cdot D_1 \cdots D_k) \equiv_w \lambda D \cdot D_1 \cdots D_k$ and $f^*(D' \cdot D_1 \cdots D_k) \equiv_w \lambda' D' \cdot D_1 \cdots D_k$ for two real numbers $\lambda \neq \lambda'$. Then $D \cdot D' \cdot D_1 \cdots D_k \not\equiv_w 0$.*

Dynamics with Hodge Index Theorem

- $n = \dim X$ and X is FT (and hence MDS by BCHM);
- $\rho = \rho(X)$, the Picard number;
- e.v. set of $f^*|_{N^1(X)}: \Lambda := \{\lambda_1, \dots, \lambda_r\}$ (real numbers after iteration);
- extremal nef divisors: $D_1, \dots, D_m$ in $\text{Nef}(X)$. ($m \geq \rho \geq r$)

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Proposition (Meng-Z., 2020)

If $r \geq n$, then the following hold.

- ① $m = \rho(X) = n = r$.
- ② $\text{Nef}(X)$ is generated by base-point-free divisors $D_1, \dots, D_n$ such that $f^*D_i \sim \lambda_i D_i$.
- ③ For each i , λ_i is a positive integer, $D_i^2 \equiv_w 0$ and $\kappa(X, D_i) = 1$.
- ④ $D_1 \cdots D_n > 0$ and $\deg f = \lambda_1 \cdots \lambda_n$.
- ⑤ $\text{Nef}(X) = \text{PE}(X)$. In particular, X is **minimal** and a Fano variety.

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- D_1, D_2 lin. indep. $\Rightarrow D_1 \cdot D_2 \not\equiv_w 0$
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 $\lambda_1 \lambda_2 \neq \lambda_1 \lambda_3 \Rightarrow D_1 \cdot D_2 \cdot D_3 \neq 0$
By projection formula, $\deg f = \lambda_1 \lambda_2 \lambda_3$.

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Proof for the case $n = \dim X = 3$ (continued)

- Suppose $m > 3$. Take D_4 s.t. $f^*D_4 \equiv \mu D_4$ and $\mu \in \{\lambda_1, \dots, \lambda_r\}$ (Note that our μ may (or may not) equal λ_i for some $i \leq 3$.)

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 $\mu\lambda_3 \neq \mu\lambda_2 \Rightarrow D_2 \cdot D_3 \cdot D_4 \neq 0$
By projection formula, $\deg f = \lambda_2\lambda_3\mu$.

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By projection formula, $\deg f = \lambda_2\lambda_3\mu$.
- Similar argument shows $\deg f = \lambda_1\lambda_3\mu$. ($\Rightarrow \Leftarrow$) (1) (2) and (4) ✓

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- The same argument shows $D_i^2 \equiv_w 0$. (3) ✓

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- The same argument shows $D_i^2 \equiv_w 0$. (3) ✓
- Suppose $D \in \partial \text{Nef}(X)$. Then WLOG, $D = a_1 D_1 + a_2 D_2$ with $a_i \geq 0$.
 $\Rightarrow (D^3 = 0) \Rightarrow (D \in \partial \text{PE}(X))$

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 $\Rightarrow (D^3 = 0) \Rightarrow (D \in \partial \text{PE}(X))$
- In particular, $-K_X$ is big $\Rightarrow -K_X$ is ample. (5) ✓

Idea for the proof of Main results

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- X smooth rationally connected; $\dim X = n$; (SRC)
- $f : X \rightarrow X$ int-amplified endomorphism. (IAP)
- the e.v. of $f^*|_{N^1(X)}$ (without counting multiplicities) are n distinct real numbers. (n -distinct real e.vs)

Induction on the dimension from two aspects

A Induction on the fibre dimension of the litaka fibration of each D_i :

$$\phi_i : X \rightarrow Y_i; \text{ (The first dynamical rigidity)}$$

B Induction on the dimension of the image of Fano contractions:

$$\pi_i : X \rightarrow X_i. \text{ (The second dynamical rigidity)}$$

- The induced morphism $\psi := (\pi_i, \phi_i) : X \rightarrow X_i \times Y_i \cong (\mathbb{P}^1)^{\times n}$

The first dynamical rigidity

- X SRC and $\dim X = n$; $f : X \rightarrow X$ surjective endomorphism.
- n distinct eigenvalues of $f^*|_{N^1(X)}$ greater than 1 ($\Rightarrow X$ F.T.)

$$\Lambda := \{\lambda_1, \dots, \lambda_n\}$$

- $f^*D_i = \lambda_i D_i$ for nef extremal divisors (BPF).

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Proposition (For the Induction A (red parts))

There exist f -periodic prime divisors $D_1, \dots, D_n$ such that:

- ① each D_i is a smooth Fano projective variety and $f^*D_i \sim \lambda_i D_i$;
- ② there exist unique (up to isomorphism) f -equivariant fibrations

$$\phi_i : X \rightarrow Y_i \cong \mathbb{P}^1$$

with $f|_{Y_i}$ being λ_i -polarized. Each D_i is a smooth fibre of ϕ_i ; and

- ③ replacing f by a positive power, $(f|_{D_i})^*|_{N^1(D_i)}$ has eigenvalues $\Lambda \setminus \{\lambda_i\}$.

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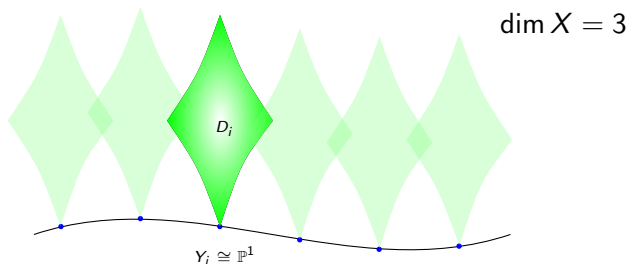
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The second dynamical rigidity

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- n distinct real eigenvalues of $f^*|_{N^1(X)}$ greater than 1

$$\Lambda := \{\lambda_1, \dots, \lambda_n\}$$

- $f^*D_i = \lambda_i D_i$ for smooth Fano projective variety D_i of dimension $n - 1$.

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Proposition (For the Induction B (red parts))

There are f -equivariant Fano contractions

$$\pi_i : X \rightarrow X_i, \quad 1 \leq i \leq n$$

of K_X -negative extremal rays, such that:

- 1 *The eigenvalues of $(f|_{X_i})^*|_{N^1(X_i)}$ are $\Lambda \setminus \{\lambda_i\}$.*
- 2 *π_i is a conic bundle and X_i is a smooth Fano projective variety.*

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- Extremal contraction of a curve C_i in $D_1 \cap \dots \cap \widehat{D_i} \cap \dots \cap D_n$.

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- Extremal contraction of a curve C_i in $D_1 \cap \dots \cap \widehat{D_i} \cap \dots \cap D_n$.
- X is minimal $\Rightarrow \pi_i := \text{Contr}_{R_{C_i}}$ is Fano contraction.

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- X is minimal $\Rightarrow \pi_i := \text{Contr}_{R_{C_i}}$ is Fano contraction.
- $K_X \cdot C_i < 0$ and $D_j \cdot C_i = 0$ for $j \neq i$.

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$$\Lambda := \{\lambda_1, \dots, \lambda_n\}$$

- $f^*D_i = \lambda_i D_i$ for smooth Fano projective variety D_i of dimension $n - 1$.

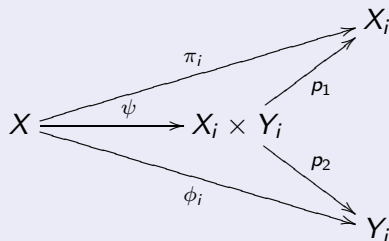
How to construct $\pi_i : X \rightarrow X_i$?

- Extremal contraction of a curve C_i in $D_1 \cap \dots \cap \widehat{D_i} \cap \dots \cap D_n$.
- X is minimal $\Rightarrow \pi_i := \text{Contr}_{R_{C_i}}$ is Fano contraction.
- $K_X \cdot C_i < 0$ and $D_j \cdot C_i = 0$ for $j \neq i$.
- Cone theorem $\Rightarrow \Lambda \setminus \{\lambda_i\}$ survive in X_i . (1) ✓

The second dynamical rigidity

sketch proofs for the second dynamical rigidity

- Easy to check ψ is finite surjective! (construct an ample divisor ψ^*H on X)

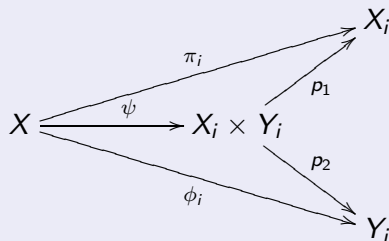


- $\dim X_i = n - 1$.

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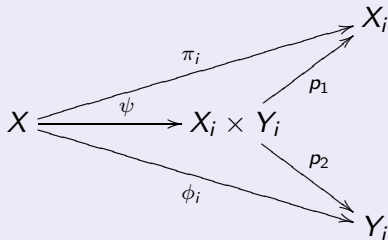


- $\dim X_i = n - 1$.
- (Ando, 1985) (equi-dimension + rel dim 1 + X smooth) $\Rightarrow X_i$ smooth & π_i a conic bundle.

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- (Ando, 1985) (equi-dimension + rel dim 1 + X smooth) $\Rightarrow X_i$ smooth & π_i a conic bundle.
- Dynamics with HIT for $(X_i, f|_{X_i}) \Rightarrow X_i$ Fano. (2) ✓

The proof for the main result

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Main result

X SRC, $\dim X = n$. Then $X \cong (\mathbb{P}^1)^{\times n} \Leftrightarrow (\text{IAP}) + (n \text{ distinct real e.vs})$.

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Proof for the direction $(\Rightarrow)$

- $X \cong Y_1 \times \cdots \times Y_n$ & $Y_i \cong \mathbb{P}^1$.
- Let $f_i : Y_i \rightarrow Y_i$ be an $(i+1)$ -polarized endomorphism:

$$f_i([a : b]) = [a^{i+1}, b^{i+1}]$$

- Take $f := f_1 \times \cdots \times f_n$.
- Easy to check $f^*|_{N^1(X)}$ has eigenvalues $\{2, 3, \dots, n+1\}$. $(\Rightarrow)$ ✓

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Proof for the direction $(\Leftarrow)$

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Proof for the direction ($\Leftarrow$)

The information we already have:

- X SRC & $\dim X = n$.
- $f : X \rightarrow X$ an int-amplified endomorphism.
- n distinct real eigenvalues of $f^*|_{N^1(X)}$ greater than 1.

$$\Lambda := \{\lambda_1, \dots, \lambda_n\}$$

- By [Yoshikawa,2020], X is FT.

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Proof for the direction $(\Leftarrow)$ (continued)

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Proof for the direction ($\Leftarrow$) (continued)

- If $n = 1$, trivial;

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Proof for the direction $(\Leftarrow)$ (continued)

- If $n = 1$, trivial;
- If $n = 2$, early proposition $\Rightarrow$ (Fano & $\rho(X) = 2$ & $\text{Nef}(X) = \text{PE}(X)$)
 $\Rightarrow$ (X smooth toric Fano surface of Picard number 2 &
 $\text{Nef}(X) = \text{PE}(X)$) $\Rightarrow$ ($X \cong \mathbb{P}^1 \times \mathbb{P}^1$).

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 $\Rightarrow$ (X smooth toric Fano surface of Picard number 2 & $\text{Nef}(X) = \text{PE}(X)$) $\Rightarrow$ ($X \cong \mathbb{P}^1 \times \mathbb{P}^1$).
- Assume our main result holds for $n - 1$ with $n \geq 3$.
- What we want?

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- If $n = 1$, trivial;
- If $n = 2$, early proposition $\Rightarrow$ (Fano & $\rho(X) = 2$ & $\text{Nef}(X) = \text{PE}(X)$)
 $\Rightarrow$ (X smooth toric Fano surface of Picard number 2 & $\text{Nef}(X) = \text{PE}(X)$) $\Rightarrow$ ($X \cong \mathbb{P}^1 \times \mathbb{P}^1$).
- Assume our main result holds for $n - 1$ with $n \geq 3$.
- What we want?

$$\psi : X \cong X_i \times Y_i \cong (\mathbb{P}^1)^{\times n}$$

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Proof for the direction $(\Leftarrow)$ (continued)

- Fix some i . **Goal:** $d := \deg \psi = l_i \cdot D_i = 1$.

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Proof for the direction $(\Leftarrow)$ (continued)

- Fix some i . **Goal:** $d := \deg \psi = l_i \cdot D_i = 1$.
- (The 1st dynamical rigidity + Induction) $\Rightarrow (\phi_i : X \rightarrow Y_i \cong \mathbb{P}^1)$ with general fibre $D_i \cong (\mathbb{P}^1)^{\times(n-1)}$.

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Proof for the direction $(\Leftarrow)$ (continued)

- Fix some i . **Goal:** $d := \deg \psi = l_i \cdot D_i = 1$.
- (The 1st dynamical rigidity + Induction) $\Rightarrow (\phi_i : X \rightarrow Y_i \cong \mathbb{P}^1)$ with general fibre $D_i \cong (\mathbb{P}^1)^{\times(n-1)}$.
- (The 2nd dynamical rigidity + Induction) $\Rightarrow (\pi_i : X \rightarrow X_i = Z_1 \times \cdots \times Z_{i-1} \times Z_{i+1} \times \cdots \times Z_n \cong (\mathbb{P}^1)^{\times(n-1)})$.
- $f|_{Z_j}$ is λ_j -polarized, when $j \neq i$.

The proof for the main result

Proof for the direction $(\Leftarrow)$ (continued)

The proof for the main result

Proof for the direction ($\Leftarrow$) (continued)

• Consider the following diagram:

- ▶ $f|_{Y_j}$ and $f|_{Z_j}$ are both λ_j -polarized.
- ▶ $X_i \cong Z_1 \times \cdots \times \widehat{Z_i} \times \cdots \times Z_n \ni (u_1, \dots, \widehat{u_i}, \dots, u_n)$. (Induction)
- ▶ $Y_j \cong Z_j \cong \mathbb{P}^1$.

$$\begin{array}{ccc} X & \xrightarrow{\pi_i} & X_i \\ \phi_j \downarrow & & \downarrow p_j \\ Y_j & & Z_j \end{array}$$

The proof for the main result

Proof for the direction ($\Leftarrow$) (continued)

- Consider the following diagram:
 - $f|_{Y_j}$ and $f|_{Z_j}$ are both λ_j -polarized.
 - $X_i \cong Z_1 \times \cdots \times \widehat{Z}_i \times \cdots \times Z_n \ni (u_1, \dots, \widehat{u}_i, \dots, u_n)$. (Induction)
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$$\begin{array}{ccc} X & \xrightarrow{\pi_i} & X_i \\ \phi_j \downarrow & & \downarrow p_j \\ Y_j & & Z_j \end{array}$$

- Uniqueness property $\Rightarrow (Z_j = Y_j) \ \& \ (p_j \circ \pi_i = \phi_j)$.

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Proof for the direction ($\Leftarrow$) (continued)

- Consider the following diagram:
 - $f|_{Y_j}$ and $f|_{Z_j}$ are both λ_j -polarized.
 - $X_i \cong Z_1 \times \cdots \times \widehat{Z_i} \times \cdots \times Z_n \ni (u_1, \dots, \widehat{u_i}, \dots, u_n)$. (Induction)
 - $Y_j \cong Z_j \cong \mathbb{P}^1$.

$$\begin{array}{ccc} X & \xrightarrow{\pi_i} & X_i \\ \phi_j \downarrow & & \downarrow p_j \\ Y_j & & Z_j \end{array}$$

- Uniqueness property $\Rightarrow (Z_j = Y_j) \ \& \ (p_j \circ \pi_i = \phi_j)$.
- $D_1 \cap \cdots \cap \widehat{D_i} \cap \cdots \cap D_n$ is a general fibre l_i of π_i .
(a general fibre of π_i is the intersection of $\phi_j^*(u_j) = D_j$ with $j \neq i$.)

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Proof for the direction $(\Leftarrow)$ (continued)

The proof for the main result

Proof for the direction ($\Leftarrow$) (continued)

- Goal: $d := \deg \psi = l_i \cdot D_i = 1$.

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- Goal: $d := \deg \psi = l_i \cdot D_i = 1$.



$$d := D_1 \cdots D_n = D_i \cdot l_i$$

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Proof for the direction ($\Leftarrow$) (continued)

- Goal: $d := \deg \psi = l_i \cdot D_i = 1$.

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- for $j \neq i$ ($D_i^2 \equiv_w 0$)

$$D_i \cdot l_j = 0$$

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$$D_i \cdot l_j = 0$$

- $-K_X$ lie in the interior of $\text{Nef}(X)$

$$-K_X \sim \sum_{i=1}^n a_i D_i$$

- Since $K_X \cdot l_i = -2$, we have $a_i = \frac{2}{d}$.

The proof for the main result

Proof for the direction $(\Leftarrow)$ (continued)

- Goal: $d := \deg \psi = l_i \cdot D_i = 1$.

The proof for the main result

Proof for the direction ($\Leftarrow$) (continued)

- Goal: $d := \deg \psi = l_i \cdot D_i = 1$.

- $a_i = \frac{2}{d} \implies$

$$-dK_X \sim 2 \sum_{i=1}^n D_i \quad (*).$$

- $(*)^{n-1} \cdot D_1:$

$$(-dK_X)^{n-1} \cdot D_1 = 2^{n-1}(n-1)!d$$

The proof for the main result

Proof for the direction ($\Leftarrow$) (continued)

- Goal: $d := \deg \psi = l_i \cdot D_i = 1$.

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$$(-dK_X)^{n-1} \cdot D_1 = 2^{n-1}(n-1)!d$$

- Applying the adjunction and noting $D_1 \cong (\mathbb{P}^1)^{\times(n-1)}$ (Induction)

$$K_{D_1} = (K_X + D_1)|_{D_1} = K_X|_{D_1} \text{ and } (-K_{D_1})^{n-1} = 2^{n-1}(n-1)!$$

- LHS of $(*)^{n-1} \cdot D_1 = 2^{n-1}(n-1)!d^{n-1}$.

- $d^{n-1} = d, n \geq 3 \Rightarrow d = 1$. ($\Leftarrow$) ✓ Done!

Further Questions

Question (Meng-Z., 2020)

Let f be an int-amplified endomorphism of SRC X . Will X split as

$$\psi : X \cong X_1 \times \cdots \times X_r$$

with each X_i is a smooth projective variety of Fano type and $f|_{X_i}$ is λ_i -polarized? (r the number of distinct real eigenvalues $< \dim X$)

Several difficulties

- $X \rightarrow X_i$ litaka fibration of D_i with $f^*D_i = \lambda_i D_i$. Smoothness of X_i ?
- litaka fibration is a trivial bundle?
- How to choose the suitable Fano contraction? (For induction)
- $\deg \psi = 1$?

Thank You!